Report R7 — Closing the monitored/unmonitored gap: the pair graph, exactness certificates, and weak charges

Krylov sector analysis of quantum cellular automata (Tier 1d)

QCA Sector Analysis project (Tier 1)

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Abstract

Report R5 established that the directed transition graph M is not a lossy picture of a dissipative rule's channel but the *exact* dynamics of a different experiment — the *monitored* QCA, in which every qubit is measured each cycle — and that for 92 of the 240 `pbc` rules the *unmonitored* channel Φ has an attractor that no set of computational-basis states can name. This report closes that gap constructively. We introduce the doubled-space *pair graph* G_2, whose diagonal is exactly M and whose cyclic structure entrywise bounds the support of every peripheral eigenoperator of Φ (Theorem 3). This yields, per (rule, N), a rigorous *certificate* (Theorem 4): when it fires, the Tier-1a attractor data is exact for the unmonitored channel. We validate the whole construction against the dense 4^N superoperator — the certificate is *sound* on all 480 dissipative units at $N = 4$ (never declaring exactness where coherence exists) and the support bound holds for every peripheral eigenoperator — and report the certificate coverage over the sweep. The pair graph also exposes a class of conservation law invisible to M: diagonal *weak charges*, which grade the protected coherence of the wall-Hadamard core (rules $22, 23, 146, 151, 178$), resolving an open item of R5. Two corrections to the naive picture are worth stating up front. First (erratum), graph-transient population does *not* generally decay under Φ: rule 22 keeps $\approx 12.5\%$ of its weight on states the monitored graph calls transient. The correct invariant is sector confinement (Theorem 1), not transience. Second, "Hadamard-free $\Rightarrow$ classical" is too coarse: a `V`-free rule with a limit cycle of period ≥ 2 carries a *protected rotating coherence* between the synchronised configurations, and correctly fails the certificate; only rules whose cycle labels de-synchronise (the additive rules 90/165) dephase it.

1 Setup and the two experiments

A dissipative rule's one-cycle channel is $\Phi(\rho) = \sum_b K_b \, \rho \, K_b^\dagger$, with the global Kraus operators K_b indexed by the frozenset b of sites at which the jump branch A_1 fired in the fixed even-then-odd brickwork order (R6, `core/cycle.py`). Two experiments share these Kraus operators:

- the **monitored** QCA measures every qubit in the computational basis each cycle. Its dynamics is the classical Markov chain $M_{yx} = \sum_b |\langle y|K_b|x\rangle|^2$ — exactly the Tier-1a transition graph.
- the **unmonitored** channel is Φ itself. Coherences feed back into populations, so Φ is *not* a function of M.

The gap is purely inter-cycle. By the one-cycle no-interference theorem (R1): starting from a basis state, a single brick-wall cycle cannot cancel any amplitude, so after one cycle the monitored and unmonitored *populations* are identical (and so is the support) — measuring at the end of cycle one changes nothing. The entire monitored/unmonitored gap is therefore a strictly multi-cycle effect: it first appears at cycle two, when a coherence created in cycle one either survives (unmonitored) and re-enters the next cycle's controls, or is dephased away (monitored). Monitoring is exactly the insertion of a computational-basis dephasing between consecutive cycles, i.e. the projection of the pair graph G_2 (§3) onto its diagonal M at every step; the off-diagonal structure of G_2 is the surviving inter-cycle coherence that this report charts.

Erratum to the design notes. An early design note asserted that population on monitored-transient states decays under Φ. This is false in general. For rule 22 (`IVVD`) at $N = 4$ the stationary states of Φ have $\approx 12.5\%$ of their weight on states that are transient for M; the attractor support is an 11-state set that strictly contains the 3 monitored pointer states. What actually holds is sector confinement, Theorem 1: a state may be transient for M yet immortal for Φ, provided it lies in the same weakly connected component as a surviving coherence. This is why the pair graph is built on the full sector block, not on the recurrent states alone.

2 Definitions

Write $d = 2^N$. Operators act on $\mathbb{C}^d$; superoperators on $\mathrm{End}(\mathbb{C}^d)$, which we coordinatise by pairs (x, y) of basis states, $A = \sum_{x,y} A_{xy}|x\rangle\langle y|$.

Definition 1 (fixed-point data)**.** $\mathrm{Fix}(\Phi) = \{A : \Phi(A) = A\}$ is the fixed-point *operator* space; its dimension is the *geometric* multiplicity of eigenvalue 1, computed by SVD (R6, `quantum/peripheral.py`). A *peripheral* eigenoperator has $|\lambda| = 1$. The Césaro projection $\overline{\Phi} = \lim_T \frac{1}{T}\sum_{t<T} \Phi^t$ exists, is itself CPTP, kills every $|\lambda| < 1$ and every peripheral phase, and has range $\mathrm{Fix}(\Phi)$; its rank is the *within-sector* $\dim \mathrm{Fix}$ we use below.

Three attractor sizes are compared throughout:

$$K_M := \#\{\text{terminal SCCs of } M\} \;\leq\; \dim \mathrm{Fix}(\Phi) \;\leq\; |P_{\mathrm{rec}}|,$$

with P_{rec} defined in §3. K_M is a purely monitored (classical) count; $\dim \mathrm{Fix}$ is the channel's exact fixed-point dimension; $|P_{\mathrm{rec}}|$ is the pair graph's combinatorial upper bound.

3 The pair graph and the theorems

Definition 2 (pair graph G_2)**.** Nodes are ordered pairs (x, y), encoded as $(x \ll N) \mid y$. There is an edge $(x, y) \to (x', y')$ iff some Kraus label b has both $\langle x'|K_b|x\rangle \neq 0$ and $\langle y'|K_b|y\rangle \neq 0$. Equivalently, joining the labelled single-cycle branches of x and of y on *identical* labels, $\mathrm{succ}_2(x, y) = \bigcup_b \mathrm{supp}(K_b|x\rangle) \times \mathrm{supp}(K_b|y\rangle)$.

The label join is exactly the same-b requirement of $\sum_b K_b(\cdot)K_b^\dagger$; joining across labels would fabricate cross-branch interference, and skipping the join would lose edges. Every matrix element is computed in the exact ring $\mathbb{Z}[1/\sqrt{2}]$, so a zero created by intra-branch interference deletes the edge exactly.

Proposition 1 (structure)**.** *(i) The diagonal subgraph $\{(x, x)\}$ of G_2 is exactly M. (ii) Swap symmetry: $(x, y) \to (x', y')$ iff $(y, x) \to (y', x')$. (iii) Confinement: every G_2-edge stays inside* $\mathrm{WCC}(x) \times \mathrm{WCC}(y)$*, where WCC denotes a weakly connected component of M.*

Parts (i)–(ii) are checked as runtime asserts (R6 tests); (iii) is Theorem 1.

Theorem 1 (sector confinement; sectors are unmonitored-exact)**.** *For a set S of basis states the following are equivalent: (a) S is closed under M-edges; (b)* $\mathrm{span}(S)$ *is invariant under every Kraus operator K_b; (c)* $\mathrm{span}(S)$ *is an enclosure of Φ. Hence the WCC partition is an* exact *invariant decomposition of the unmonitored channel, and Φ is block-diagonal across the induced operator blocks $S_i \otimes S_j^*$. What does* not *transfer is the recurrent/transient split and the SCC refinement inside a WCC: those are monitored-relative.*

Proof sketch. (a)⇔(b): $M_{yx} \neq 0$ iff some K_b has $\langle y|K_b|x\rangle \neq 0$; closure of S under all such y is exactly invariance of $\mathrm{span}(S)$ under each K_b (a one-step, interference-immune property). (b)⇒(c): if each K_b preserves $\mathrm{span}(S)$ then so does Φ, and by trace preservation no weight leaks out. Block-diagonality follows because $K_b|x\rangle\langle y|K_b^\dagger$ keeps the left index in $\mathrm{WCC}(x)$ and the right in $\mathrm{WCC}(y)$. □

Theorem 2 (lower bound)**.** $K_M \leq \dim \mathrm{Fix}(\Phi)$*: each terminal SCC carries at least its own stationary state.*

Theorem 3 (upper bound)**.** *Let $P_{\mathrm{rec}} := \mathrm{FwdCl}_{G_2}$(cyclic pairs of G_2), where a* cyclic pair *lies in a G_2-SCC of size > 1 or carries a self-loop. Then every peripheral eigenoperator A of Φ (in particular every fixed point) has* $\mathrm{supp}(A) \subseteq P_{\mathrm{rec}}$. *Consequently* $\dim \mathrm{Fix}(\Phi) \leq |P_{\mathrm{rec}}|$*, and any protected coherence lives on the off-diagonal part of P_{rec}.*

Proof sketch. For $|\lambda| = 1$, $A = \lambda^{-t}\Phi^t(A)$ for all $t \geq 0$, so $\mathrm{supp}(A)$ is invariant under the support map of Φ, which is precisely forward propagation along G_2. A pair in $\mathrm{supp}(A)$ therefore admits arbitrarily long in-paths within $\mathrm{supp}(A)$; by finiteness (pigeonhole) it is reachable from a G_2-cycle, i.e. it lies in FwdCl of the cyclic pairs. □

The forward-closure form is the price of a purely combinatorial bound: it cannot detect amplitude cancellation, so P_{rec} can strictly overcount. It is, however, rigorous, and it is what makes the certificate below sound.

Theorem 4 (exactness certificate)**.** *If the off-diagonal part of P_{rec} is empty and its diagonal part equals the M-recurrent set, then every peripheral eigenoperator of Φ (restricted to each sector) is basis-diagonal and supported on the recurrent classes. In that case the Tier-1a attractor data (K_M, sizes) is* exact *for the unmonitored channel; we record* `certified=true`.

What the certificate does *not* claim. It is a per-sector (diagonal-block) statement. Cross-sector coherences — e.g. a stationary $|x\rangle\langle y|$ between two frozen states in *different* sectors, which inflate the full $\dim \mathrm{Fix}$ by up to F^2 for F frozen states — are the "noiseless pairing" content and are reported separately (the `--cross-sector` pass); they do not bear on the attractor *structure*. Accordingly we compare like-for-like: the sandwich uses the *within-sector* $\dim \mathrm{Fix}$ (sum of per-WCC Césaro ranks), for which $K_M \leq \dim \mathrm{Fix}_{\mathrm{within}} \leq |P_{\mathrm{rec}}|$ holds exactly.

Computation. Because a cyclic pair projects to an M-cycle on each coordinate and terminal SCCs are M-closed, P_{rec} is contained in $\mathrm{Cyc} \times \mathrm{Cyc}$ (states on M-cycles) and its forward closure; the pair graph is therefore run on the WCC-diagonal blocks only, streamed and never materialised beyond the block. Fragmented rules have small WCCs, so G_2 stays cheap far beyond the dense-ρ ceiling $N = 8$ of the Tier-1b census; near-ergodic rules are capped by a node budget and reported bound-only. (R6, `graph/pair_graph.py`.)

4 Validation

The certificate and the bound were checked against the exact 4^N superoperator (`quantum/peripheral.py`) on *all* 480 dissipative (rule, bc) units at $N = 4$ and a sample at $N = 5$:

- **Bound chain:** $K_M \le \dim \mathrm{Fix} \le |P_{\mathrm{rec}}|$ holds on every unit.
- **Support containment (Thm 3):** every peripheral eigenoperator of the full superoperator has entrywise support inside the cross-sector P_{rec} — 0 violations.
- **Soundness (Thm 4):** no unit certifies `true` while carrying within-sector coherence — 0 unsound certificates, checked against the WCC-restricted superoperator on all 240 dissipative rules at $N = 4$.

The certificate is *conservative*: a combinatorial support bound cannot see amplitude cancellation, so 89 of 480 genuinely diagonal units at $N = 4$ fail to certify. This is by design — when the certificate declines we report the sandwich interval, not a false negative.

A refinement of "`V`-free ⇒ classical". It is tempting to expect every Hadamard-free rule to certify. It does not, and the pair graph explains why exactly. A `V`-free rule is deterministic on basis states, so a limit cycle of period $p \ge 2$ is a genuine classical cycle $c_0 \to c_1 \to \cdots \to c_0$. The coherence $|c_i\rangle\langle c_j|$ is a peripheral eigenoperator *iff* the Kraus labels are synchronised around the cycle. Rule 10 (`DEDD`) has a 2-cycle with matched labels and therefore a protected $\lambda = \pm 1$ coherence — confirmed directly in the superoperator spectrum — so it correctly certifies `false`. The additive rules 90/165 (`DEED`, `EDDE`) have longer cycles whose reset labels de-synchronise; the coherence dephases and they certify `true` despite having a cycle. Thus the sharp criterion is label synchronisation, which G_2 computes, not the mere presence of a cycle.

5 The visibility theorem and weak charges

Theorem 5 (visibility). *Diagonal* strong *charges s : basis $\to \mathbb{Q}$ — conserved along every single-branch M-edge, equivalently constant on WCCs — are exactly the diagonal part of the strong commutant $\{K_b\}'$; their number is the diagonal dimension of that commutant. Diagonal* weak *charges impose* no *constraint on M (a diagonal unitary acts trivially on diagonal states); they are precisely the G_2 difference-conservation laws $s(x) - s(y) = const$ along every G_2-edge $(x, y) \to (x', y')$. Non-diagonal symmetries of either kind are visible only at operator level (Tier 1b / Tier 2).*

We detect both with a range-r local-density ansatz with even/odd sublattice split, solving the linear systems exactly over $\mathbb{Q}$ (R6, `quantum/weak_charges.py`). The charge spaces nest $C \subseteq S \subseteq Z$ (constants $\subseteq$ strong $\subseteq$ G_2-conserved), giving certified integer counts $n_{\mathrm{strong}} = \dim S - \dim C$ and, restricting the difference law to the off-diagonal part of P_{rec} (so that structurally-decaying coherences do not contribute vacuous laws), a count of weak charges that *grade* the protected coherence.

Result (R5 open item i). The protected coherence of the wall-Hadamard core is graded by a single weak charge, invisible to the monitored dynamics, with a small integer D-spectrum:

W	tuple	#strong	#weak	grades coherence (D-values)
22	`IVVD`	0	1	yes $\{-2, \ldots, 2\}$
23	`EVVD`	0	1	yes $\{-1, 0, 1\}$
146	`DVVI`	0	1	yes $\{-2, \ldots, 2\}$
151	`EVVI`	0	1	yes $\{-2, \ldots, 2\}$
178	`DVVE`	0	1	yes $\{-1, 0, 1\}$
18, 19, 50, 55, 179	—	0	0	no

The graded quantity is a domain-wall–like density: the weak charge partitions the off-diagonal support of P_{rec} into Δ-sectors, exactly the structure a monitored observer cannot see. The effect is not confined to the wall core: 58 of the 92 coherent `pbc` rules carry at least one weak charge that grades their protected coherence, with the wall-Hadamard core the cleanest (single charge, small symmetric Δ-spectrum).

6 Results

Certificate coverage. Figure 1 reports the fraction of dissipative units certified exact for the unmonitored channel at each N. Of the 240 `pbc` rules, 62–80 certify per size ($62, 67, 80, 67, 62, 67$ at $N = 4, \ldots, 9$; the peak at $N = 6$ is a commensurability effect). The Hadamard-free rules with only fixed-point attractors certify at every N; the 92 coherent-core rules never certify (correctly); the conservative middle is the combinatorial slack of Theorem 3.

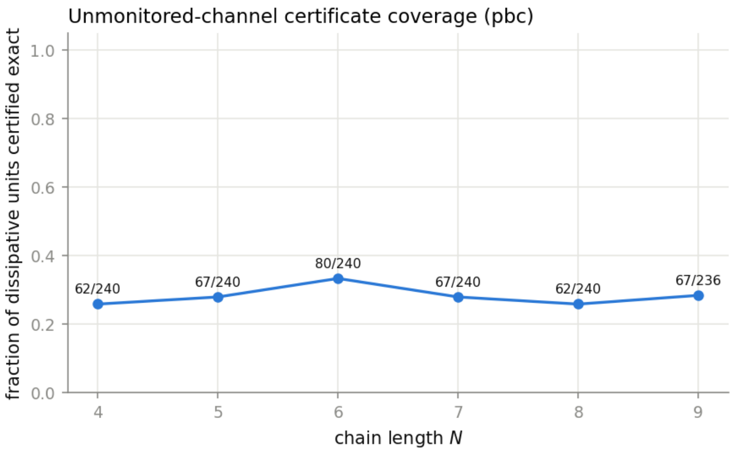

Figure 1: Unmonitored-channel certificate coverage vs. N (`pbc`).

The same verdict, laid over the Tier-1c dissipative growth-rate plane, gives the corrected growth map (Figure 2): each rule's monitored growth exponents are drawn filled when the channel is certified exact and as a hollow ring when it carries protected coherence.

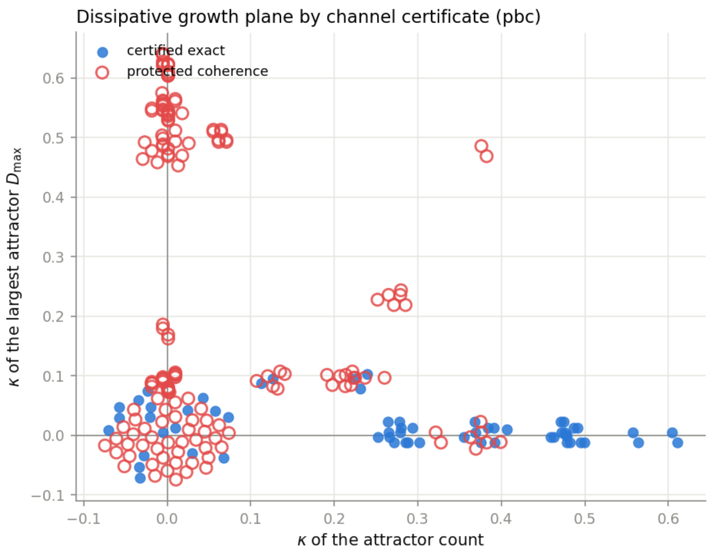

Figure 2: Dissipative growth-rate plane coloured by the channel certificate: filled markers are certified exact for the unmonitored channel, hollow rings carry protected coherence (`pbc`).

The coherent-core sandwich. For each coherent rule the three series $K_M(N) \le \dim \mathrm{Fix}_{\mathrm{within}}(N) \le |P_{\mathrm{rec}}|(N)$ are fitted with the Tier-1c machinery (exact integer recurrence first, then the two-parameter exponential, period split active); the pair-graph series live in the doubled space, so their bases run up to 4, not the base-2 ceiling of single-space counts. All 92 coherent-core rules return an *interval* verdict: the monitored lower base and the channel upper base genuinely differ (e.g. rule 22: [1.0, 3.66] with the exact within-sector anchor at 2.44), so no single graph number is tight — the honest output is a growth window, and the combinatorial upper bound is loosest exactly where interference cancellation is strongest (rule 27: [1, 3.06] for a one-dimensional stationary superposition). Table 1 lists the bracketing bases and the per-rule verdict; Figure 3 draws the intervals.

W	tuple	b_{K_M}	b_{Fix}	$b_{\lvert P_{\mathrm{rec}}\rvert}$	verdict	cert	weak
22	`IVVD`	1.0000	2.4375	3.6627	interval	×	yes
23	`EVVD`	1.0000	1.6667	2.9964	interval	×	yes
146	`DVVI`	1.0000	1.6125	3.5894	interval	×	yes
151	`EVVI`	1.0000	2.4375	3.6627	interval	×	yes
178	`DVVE`	1.0000	1.6667	2.9964	interval	×	yes
50	`DVVV`	1.0000	0.5547	3.5273	interval	×	yes
27	`VEVD`	1.0000	1.0000	3.0554	interval	×	–
18	`DVVD`	1.0000	1.0000	3.0081	interval	×	yes
19	`VVVD`	1.0000	0.9199	3.5273	interval	×	–

Table 1: Coherent-core growth-base sandwich (`pbc`): lower base b_{K_M}, exact within-sector b_{Fix} ($N \le 6$), upper $b_{|P_{\mathrm{rec}}|}$, the verdict, the certificate at $N_{\max}$, and whether a weak charge grades the coherence.

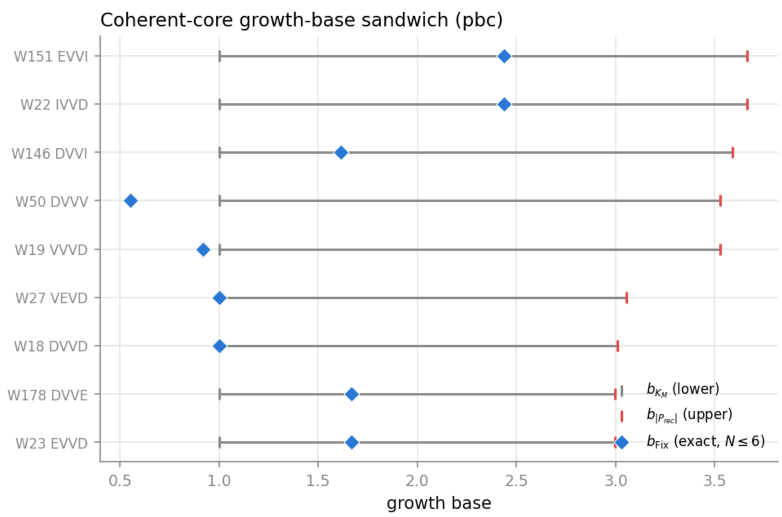

Figure 3: The growth-base sandwich for the coherent-core anchors: the whisker runs from the monitored lower base to the channel upper base, with the exact within-sector fixed-point base marked inside it.

The census non-monotonicity (R5 open item ii). The Tier-1b coherence census ran only at even N ($84, 60, 72$ at $N = 4, 6, 8$) and looked non-monotonic. The pair graph adds two things the even-only dense probe could not. First, coherence *support* (a non-empty off-diagonal P_{rec}) persists at *every* N, including odd N: $178, 173, 160, 173, 178, 169$ rules of 240 carry it at

$N = 4, \dots, 9$. So the apparent dip is not coherence vanishing and reappearing — it is present throughout. Second, the *exact* within-sector count ($\dim \mathrm{Fix}_{\mathrm{within}} > K_M$, reachable exactly to $N = 6$) is $104, 76, 58$ at $N = 4, 5, 6$; the odd point $N = 5$ sits between its even neighbours, so this exact quantity is monotone over the sizes we can certify, with no parity turnover. The specific dense-census upturn at $N = 8$ lives in a different quantity (basis-non-spannedness) and at a size the exact within-sector probe cannot reach; what we can state rigorously is that it is not a disappearance of protected coherence. Figure 4 overlays the exact even/odd counts on the dense census.

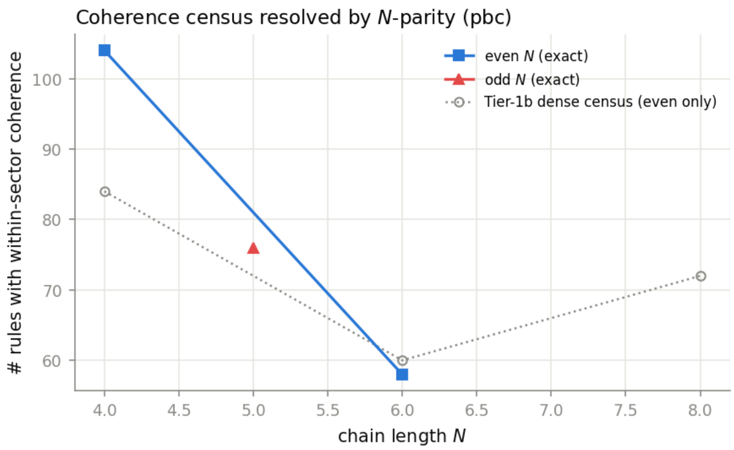

Figure 4: Protected-coherence census resolved by N-parity (`pbc`).

One-line remark on the classical family. For a `V`-free rule the Kraus operators are partial permutations, so the monitored and unmonitored *populations* coincide; the nontrivial content is entirely the coherence sharpness among the `V`-carrying rules, and — as §4 shows — even the classical family is not uniformly certifiable once limit cycles appear.

7 Outlook

The pair graph is the $p \to \infty$ end (measure every cycle is M; never measure is Φ) of a measurement-every-p-cycles family; the intermediate Zeno axis connects to measurement-induced phase transitions. The weak-charge candidates and the diagonal/off-diagonal P_{rec} structure are the natural handoff to a Tier-2 commutant analysis: a weak charge is a diagonal element of the bilocal commutant that the monitored chain cannot see, and its Δ-grading of P_{rec} is a concrete decomposition for the operator-level symmetry search.

Reproduce.

```
python -m qca_fragmentation.run_rule --rule 22 --N 4-10 --bc pbc --tier 1d
python -m qca_fragmentation.pair_sweep --which coherent --n-min 4 --n-max 12
python -m qca_fragmentation.quantum.weak_charges --core
python -m qca_fragmentation.scaling.pair_scaling --bc pbc
python -m qca_fragmentation.scaling.pair_figure --bc pbc
python -m pytest qca_fragmentation/tests/test_pair_graph.py -q
```